

Math 141, S06
Exam 1

Section:

Name: *key*
SSN: xxx-xx

Instructions:

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	18	
2	16	
3	16	
4	24	
5	10	
6	16	
Total	100	

Good Luck !

Problem 1. (18 points) Find the natural domain of each of the following functions.

(a) $f(x) = \frac{x-3}{x^2-9}$

Solution: We need to have.

$$x^2 - 9 \neq 0$$

$$x^2 \neq 9$$

$$x \neq 3 \text{ or } -3.$$

So the natural domain of f is

$$(-\infty, -3) \cup (-3, 3) \cup (3, +\infty)$$

(b) $g(x) = \sqrt{x-4x^2}$

Solution: We need to make

$$x - 4x^2 \geq 0$$

$$4x^2 - x \leq 0$$

$$x(4x-1) \leq 0$$

$$0 \leq x \leq \frac{1}{4}$$

(c) $h(\theta) = \frac{1}{1 - \cos \theta}$

Solution. We need to have

$$1 - \cos \theta \neq 0$$

$$\cos \theta \neq 1$$

$$\theta \neq 2k\pi \text{ for } k=0, \pm 1, \pm 2, \pm 3, \dots$$

Problem 2. (16 points) Consider the parametric equations $x = t - 4$ and $y = 2 - \sqrt{t}$ with $t \geq 0$.

(a) Express this curve in the form of $y = f(x)$.

Solution: $x = t - 4$

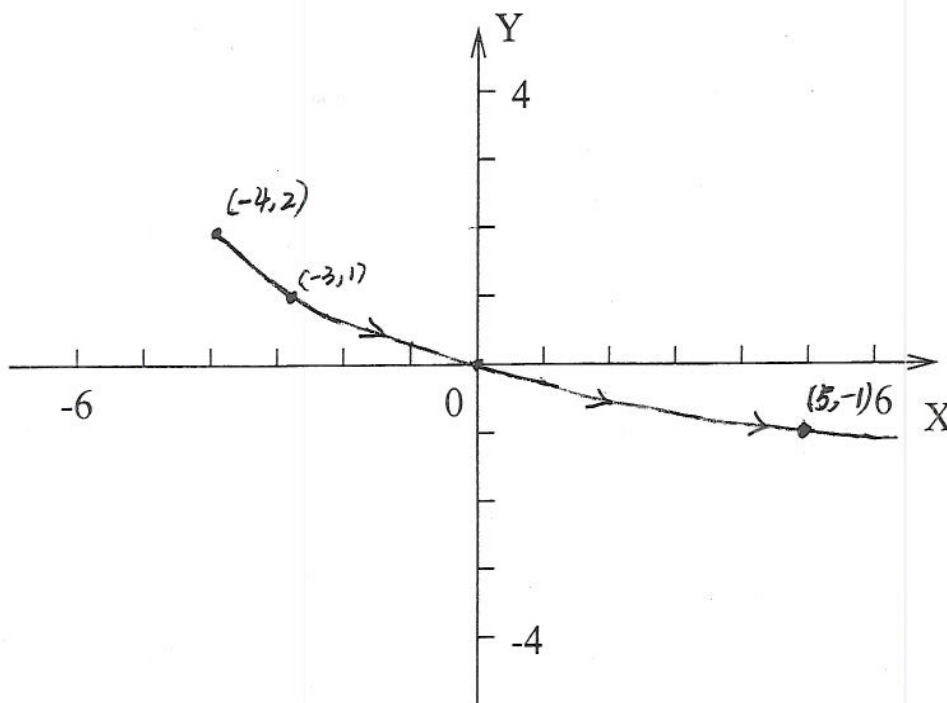
$$t = x + 4$$

Then,

$$y = 2 - \sqrt{t}$$

$$y = 2 - \sqrt{x+4}$$

(b) Sketch this curve in the viewing window $[-6, 6] \times [-4, 4]$ using the result of (a) and indicate the direction of increasing t .



Problem 3. (16 points)

(a) Simplify

$$\begin{aligned} & 3^2 \cdot 3^{-1} + 27^{-\frac{1}{3}} + \log_4 2 + \log_4 8 + e^{\frac{1}{2} \ln 9} \\ &= 3^{2-1} + \frac{1}{27^{\frac{1}{3}}} + \log_4 (2 \cdot 8) + e^{\ln(9^{\frac{1}{2}})} \\ &= 3 + \frac{1}{3} + \log_4 16 + e^{\ln 3} \\ &= 3 + \frac{1}{3} + 2 + 3 \\ &= 8\frac{1}{3} \end{aligned}$$

(b) Find $x > 0$ which satisfies $\ln x^2 - \ln 3x = 3$.

Solution: $\ln x^2 - \ln 3x = 3$

$$\ln\left(\frac{x^2}{3x}\right) = 3$$

$$\ln \frac{x}{3} = 3$$

$$\frac{x}{3} = e^3$$

$$x = 3e^3$$

Problem 4. (24 points) Find the following limits.

(a) $\lim_{x \rightarrow 2} \frac{x^3 + 4}{8 - x^2}$

$$= \frac{2^3 + 4}{8 - 2^2} = \frac{12}{4} = 3$$

(b) $\lim_{x \rightarrow -2} \frac{x^2 - 4}{x^2 - x - 6}$

$$= \lim_{x \rightarrow -2} \frac{(x-2)(x+2)}{(x-3)(x+2)}$$

$$= \lim_{x \rightarrow -2} \frac{x-2}{x-3} = \frac{-2-2}{-2-3} = \frac{4}{5}$$

(c) $\lim_{x \rightarrow +\infty} \frac{x^3 - 9}{4x^3 + x - 6}$

$$= \lim_{x \rightarrow +\infty} \frac{\frac{x^3 - 9}{x^3}}{\frac{4x^3 + x - 6}{x^3}} = \lim_{x \rightarrow +\infty} \frac{1 - \frac{9}{x^3}}{4 + \frac{1}{x^2} - \frac{6}{x^3}}$$

$$= \frac{1 - 0}{4 + 0 - 0} = \frac{1}{4}$$

(d) $\lim_{x \rightarrow 0^-} \frac{1}{\sin x}$

Since $\lim_{x \rightarrow 0^-} \sin x = 0$ and $\sin x < 0$ when x approaches 0 close enough from the left, we have

$$\lim_{x \rightarrow 0^-} \frac{1}{\sin x} = -\infty$$

Problem 5. (10 points) Find $\lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{x \tan x}$.

$$\begin{aligned}
 \text{Solution: } \lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{x \tan x} &= \lim_{x \rightarrow 0} \frac{\sin^2 x}{x \tan x} \\
 &= \lim_{x \rightarrow 0} \frac{\sin^2 x}{x} \cdot \frac{\cos x}{\sin x} \\
 &= \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \cos x \\
 &= \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \lim_{x \rightarrow 0} \cos x \\
 &= 1 \cdot 1 = 1
 \end{aligned}$$

Problem 6. (16 points) Let $f(x) = \frac{x^2 - 4}{|x| - 2}$. Find values of x , if any, at which the function $f(x)$ is not continuous, and determine whether each such value is a removable discontinuity.

Solution: The function $f(x)$ is not continuous at points where $|x| - 2 = 0$

$\Leftrightarrow f$ is not continuous at $2, -2$.

i) When $x = 2$,

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{|x| - 2} = \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{x-2} = \lim_{x \rightarrow 2} x+2 = 4$$

So it is a removable discontinuity

ii) When $x = -2$,

$$\lim_{x \rightarrow -2} \frac{x^2 - 4}{|x| - 2} = \lim_{x \rightarrow -2} \frac{(x-2)(x+2)}{-x-2} = \lim_{x \rightarrow -2} -(x-2) = 4$$

So it is still a removable discontinuity.